

30. Answer: B

$$A = \lambda N$$

$$A = \frac{\ln 2}{t_{\frac{1}{2}}} N$$

$$\Rightarrow A_x = \frac{\ln 2}{t_x} N_x \Rightarrow t_x = \frac{\ln 2}{A_x} N_x \text{----- (1)}$$

$$\Rightarrow A_y = \frac{\ln 2}{t_y} N_y \Rightarrow t_y = \frac{\ln 2}{A_y} N_y \text{----- (2)}$$

$$\begin{aligned} \frac{t_x}{t_y} &= \frac{N_x}{A_x} \times \frac{A_y}{N_y} \\ &= \frac{4.00 \times 10^{19}}{2.00 \times 10^{19}} \times \frac{4.60 \times 10^7}{3.68 \times 10^8} \\ &= 0.25 \end{aligned}$$